

Key Concepts in AI-Driven Assistive Robotics

This flyer describes five key concepts you should know as a beginner in artificial intelligence-driven assistive robotics, as well as why these concepts are important.

01

Artificial Intelligence (AI)

The ability for machines and computers to perform human-level tasks such as decision-making, learning, and predicting. AI is important because it allows assistive robots to make decisions based on previous data.

02

Human-Robot Interaction (HCI)

The way humans and robots work together to complete tasks effectively. HCI is important because assistive robots must be intuitive, useful, and accessible for those in need.

03

Sensors

Devices that take in physical data, such as motion or sounds, and send the information to the processors of the robot. Sensors are important because they provide the data to the artificial intelligence system, which tells the robot how to assist the user in need.

04

Design for Accessibility

The design of products that can be used by people of all abilities, backgrounds, experience levels. Designing for accessibility is important because it promotes inclusivity in society and makes daily life less challenging for those in need.

05

Assistive Robotics

A branch of robotics that focuses on robots that assist people with disabilities, injuries, or other challenges. Assistive robotics is important because it improves the day-to-day experience of those in need in an autonomous and personalized fashion.

References: 1. "AI and Robotics: How Artificial Intelligence is Transforming Robotic Automation" , Encord Blog, Haziqa Sajid

2. "Human-Robot Interaction: Key Concepts & What Lies Ahead" , BOTA Systems, Martin Wermelinger